Probability and Statistics
Unit-VI
Curve fitting and Regression

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UNIT-VI

Curve fitting and Regression:

Curve fitting by the method of least squares- fitting of straight lines, second degree parabolas and more general curves. Types of Regression, linear regression, multiple regressions.

Introduction:

With an experimental data, the data is plotted on a graph paper and a straight line is drawn through the plotted points. This is the usual method to fit a mathematical equation to experimental data. The method of least squares is the most systematic procedure to fit a unique curve through the given data points.

Its application is wide in practical computations.

Let the data points be (x_i, y_i), $i=1,2,3\ldots m$. Suppose the curve $y=f(x)$ is fitted to this data. Let the observed value at $x=x_i$ is y_i and the corresponding value on the curve is $f(x_i)$. Let e_i be the error of approximation at $x=x_i$, then we have

$e_i=y_i-f(x_i)$

Consider,

The method of least squares consists of minimizing S.

Fitting a Straight line:

Let $y = a_0+a_1x$ is a straight line to be fitted to the given data.

The normal equations are

Here N= No.of data points.

Solving the above two equations we get a_0 and a_1 and hence we the required fitting a straight line

$y = a_0+a_1x$.

I. Fitting a straight line :-

Let $y = a_0 + a_1 x$ be a st. line to be fitted to the given data. then the normal eq are :-

$$\Sigma y = a_0 N + a_1 \Sigma x \quad \text{—— (1)}$$

$$\Sigma xy = a_0 \Sigma x + a_1 \Sigma x^2 \quad \text{—— (2)}$$

solving (1) & (2)

we get a_0 & a_1.

where 'N' = the no. of data Points.

Problems:-

(1) By the method of least squares, find the st. line that best fits the following data and estimate the value of "y" corresponding to "$x = 7$".

x	1.	2	3	4.	5
y	14	27	40	55	68

sol:- let $y = a_0 + a_1 x$ — (1) be the st. line to fit to given data.

the normal eq are

x	x^2	y	xy
1	1	14	14
2	4	27	54
3	9	40	120
4	16	55	220
5	25	68	340
$\Sigma x = 15$	$\Sigma x^2 = 55$	$\Sigma xy = 204$	$\Sigma xy = 748$

from eq 1 & 2

$\Sigma x = 15$, $\Sigma x^2 = 55$, $\Sigma y = 204$, $\Sigma xy =$

$N = 5$

$$204 = 5a_0 + 15a_1 \quad \text{--- (3)}$$

$$748 = 15a_0 + 55a_1 \quad \text{--- (4)}$$

solving (3) & (4) we get

$$\boxed{a_0 = 0}, \boxed{a_1 = 13.6}$$

∴ the required st. line

$$\boxed{y = 13.6x}$$

when $x = 7$.

$$\boxed{y = 95.2}$$

(2) By the method least squares, find the best fitting st. line to the data given below.

x	5	10	15	20	25
y	15	19	23	26	30

Sol:-

x	x^2	y	xy
5	25	15	75
10	100	19	190
15	225	23	345
20	400	26	520
25	625	30	750
$\Sigma x = 75$	$\Sigma x^2 = 1375$	$\Sigma y = 113$	$\Sigma xy = 1880$

$N = 5$.

$113 = a_0 5 + 75 a_1$ — (3)

$1880 = 75 a_0 + 1375 a_1$ — (4)

$\boxed{a_0 = 11.5}$, $\boxed{a_1 = 0.74.}$

the required st. line.

$\therefore \boxed{y = 11.5 + 0.74x.}$

(3) fit the least square st. line $y = a + bx$ to the following data.

x	-5	3	-1	1	2	4
$f(x)$	0.4	-0.1	-0.3	-0.3	0.1	0.4

(4) the temp (T) (in °c) and lens L (in mm) of a heated rod are given below. If $L = a_0 + a_1 T$ find the best value for a_0 and a_1

T	20	30	40	50	60	70
L	800.3	800.4	800.6	800.7	800.9	800.1

Normal eq:-

$$\Sigma L = a_0 N + a_1 \Sigma T \quad \text{— (1)}$$

$$\Sigma L T = a_0 \Sigma T + a_1 \Sigma T^2 \quad \text{— (2)}$$

$\boxed{a_0 = 799.99}$, $\boxed{a_1 = 0.0146.}$

Fitting of second Degree Curve or Parabola:

suppose

fitting of 2nd degree curve (or) Parabola:

suppose the curve to be fitted with the given data is 2nd degree curve or parabola is

$$y = a_0 + a_1 x + a_2 x^2$$

the normal eq are:-

$$\boxed{\begin{aligned} \Sigma y &= a_0 N + a_1 \Sigma x + a_2 \Sigma x^2 \quad \text{--- (1)} \\ \Sigma xy &= a_0 \Sigma x + a_1 \Sigma x^2 + a_2 \Sigma x^3 \quad \text{--- (2)} \\ \Sigma x^2 y &= a_0 \Sigma x^2 + a_1 \Sigma x^3 + a_2 \Sigma x^4 \quad \text{--- (3)} \end{aligned}}$$

solving ①, ②, ③ we get.

$\simeq a_0, a_1, a_2, a_3$

Problem :-

① fit a 2^{nd} degree polynomial to the following data by the method of least square :

x	10	12	15	23	20
y	14	17	23	25	21

Sol:-

x	x^2	x^3	x^4	xy	y	x^2y
10	100	1000	10000	140	14	1400
12	144	1728	20736	204	17	2448
15	225	3375	50625	345	23	5175
23	529	12167	279841	575	25	13225
20	400	8000	160000	420	21	8400
$\Sigma x = 80$	$\Sigma x^2 = 1398$	$\Sigma x^3 = 26270$	$\Sigma x^4 = 581207$	$\Sigma xy = 1684$	$\Sigma y = 100$	$\Sigma x^2y = 30648$

$N = 5$,

$\Sigma x = 80$, $\Sigma x^2 = 1398$, $\Sigma x^3 = 26270$, $\Sigma x^4 = 521202$.

$\Sigma xy = 1684$, $\Sigma y = 100$, $\Sigma x^2 y = 30648$

from eq. (1) & (2) & (3)

$$100 = 5a_0 + 80a_1 + 1398a_2 \quad \text{—} (4)$$

$$1684 = 80a_0 + 1398a_1 + 26270a_2 \quad \text{—} (5)$$

$$30648 = 1398a_0 + 26270a_1 + 521202a_2 \quad (6)$$

$a_0 = -8.72$, $a_1 = 3.0099$, $a_2 = -0.06$

the required eq. is :-

$y = -8.72 + 3.0099x + (-0.0697)x^2$

$y = -8.72 + 3.0099x - 0.069x^2$

$\Rightarrow \boxed{y = -8.72 + 3.0099x - 0.069x^2}$

(2) find the Parabola of the form $y = ax^2 + bx + c$ Passing through the Points $(-1, 2)$, $(0, 1)$, $(1, 4)$ using method of least squares

Sol:- given Parabola,

$$y = c + bx + ax^2 \quad \text{—— } (*)$$

the corresponding normal eq:-

$$\Sigma y = cN + b\Sigma x + a\Sigma x^2 \quad \text{—— } (1)$$

$$\Sigma xy = c\Sigma x + b\Sigma x^2 + a\Sigma x^3 \quad \text{—— } (2)$$

$$\Sigma x^2 y = c\Sigma x^2 + b\Sigma x^3 + a\Sigma x^4 \quad \text{—— } (3)$$

$$\boxed{N = 3}$$

x	y	x^2	x^3	x^4	xy	x^2y
-1	2	1	-1	1	-2	2
0	1	0	0	0	0	0
1	4	1	1	1	4	4
$\Sigma x=0$	$\Sigma y=7$	$\Sigma x^2=2$	$\Sigma x^3=0$	2	2	6

$$7 = c(3) + b(0) + a(2) \quad \text{—} \quad (4)$$

$$2 = c(0) + b(2) + a(0) \quad \text{—} \quad (5)$$

$$6 = c(2) + b(0) + a(2) \quad \text{—} \quad (6)$$

$c = 1, b = 1, a = 2$.

Hence :-

$$\boxed{y = 1 + x + 2x^2}$$

(3) fit a 2^{nd} degree polynomial to the following data by the method of least squares.

x	0	1	2	3	4
y	1	1.8	1.3	2.5	6.3

A: $(a_0 = 1.42,\ a_1 = -1.07,\ a_2 = 0.55)$

x	0	1	2
y	1	6	17

$(a_0 = 1,\ a_1 = 2,\ a_2 = 3)$

III) fitting of Exponential curve :-

(i) suppose the curve to be fitted in the given data is $\boxed{y = ae^{bx}} \longleftarrow (1)$

log on both side.

$$\log_e y = \log_e a + \log_e e^{bx}$$

$$= \log_e a + bx \cdot \log_e e$$

$\log_e y = \log_e a + bx$

$$\boxed{Y = A + Bx} \quad \text{--- (2)}$$

where $Y = \log_e y$, $A = \log_e a$, $B = b$, $X = x$.

fitting of exponential curve is reduced to fitting of st. line.

The corresponding normal eq for eq (2) are:-

$$\boxed{\begin{aligned} \Sigma Y &= AN + B\Sigma X \\ \Sigma XY &= A\Sigma X + B\Sigma X^2 \end{aligned}}$$

(ii) let the exponential curve be

$$\boxed{y = ab^x} \quad \text{--- (3)}$$

where a, b are constant.

log on both side :-

$$\log_{10} y = \log_{10}(ab^x)$$

$$\log_{10} y = \log_{10} a + \log_{10} b^x$$

$$\log_{10} y = \log_{10} a + x \log_{10} b$$

$$\Rightarrow \boxed{Y = A + Bx} \quad \text{--- (4)}$$

where, $Y = \log_{10} y$, $A = \log_{10} a$, $B = x \log_{10} b$

then fitting exponential curve is reduced to fitting of st. line.

normal eq for eq (3). are :-

$$\boxed{\begin{aligned} \Sigma Y &= AN + B\Sigma X \\ \Sigma XY &= A\Sigma X + B\Sigma X^2 \end{aligned}}$$

Problem:-

(1) determine the constant A & B by the method of least squares. such that $y = ae^{bx}$

x	2	4	6	8	10
y	4.077	11.084	30.128	81.897	222.62

sol:- given eq is

$$y = ae^{bx} \text{ --- } (*)$$

log. B.S.

$$\log_e y = \log_e a + bx$$

$$\Rightarrow \boxed{Y = A + BX} \text{ --- } (1)$$

where $Y = \log_e y$, $A = \log_e a$, $B = b$, $X = x$.

the normal eq for eq (1) :-

$$\Sigma Y = AN + B\Sigma X \text{ --- } (2)$$

$$\Sigma XY = A\Sigma X + B\Sigma X^2 \text{ --- } (3)$$

$X = x$	X^2	$Y = \log_e y$	XY	y
2	4	1.405	2.810	4.077
4	16	2.405	9.620	11.084
6	36	3.405	20.430	30.128
8	64	4.405	35.24	81.897
10	100	5.405	54.04	222.62
$\Sigma x = 30$	$\Sigma x^2 = 220$	$\Sigma Y = 17.025$	$\Sigma xY = 122.15$	

<u>normal eq :-</u>

$N = 5$.

$17.025 = 5A + 30B$ — (3)

$122.15 = 30A + 220B$ — (4)

$A = 0.405$, $B = 0.5$.

But,

$A = \log_e a$, $B = b$ $\boxed{e = 2.718}$

$0.405 = \log_e a$, $\boxed{b = 0.5}$

$\Rightarrow a = e^{0.405}$

$a = (2.718)^{0.405}$

$\Rightarrow \boxed{a = 1.499}$

from (*), $y = ae^{bx}$

$\boxed{y = 1.499\, e^{0.5x}}$

(2) find the curve of best fit of the type $y = ae^{bx}$ to the following data by the method of least squares

x	1	2	7	9	12
y	10	15	12	15	21

sol:-

$X = x$	X^2	y	$Y = \log_e y$	XY
1	1	10	2.302	2.302
2	4	15	2.708	5.416
7	49	12	2.484	17.388
9	81	15	2.708	24.372
12	144	21	3.044	36.528
$\Sigma X = 31$	$\Sigma X^2 = 279$		13.246	86.006

$N = 5.$

$$13.246 = 5A + 31B \quad \text{—— (1)}$$

$$86.006 = 31A + 279B \quad \text{—— (2)}$$

$$A = 2.372, \quad B = 0.044$$

$$A = \log_e a, \quad \boxed{B = b} \Rightarrow b = 0.044$$

$$2.372 = \log_e a.$$

$$a = e^{2.372}$$

$$\Rightarrow \boxed{a = 10.71}$$

from (*) $\Rightarrow y = ae^{bx}$

$$\boxed{y = 10.71e^{0.044x}}$$

(Q1) Q-type :-

(3) Obtain a relation of the form $y = ab^x$ for the following data by the method of least squares.

x	2	3	4	5	6
y	8.3	15.4	33.1	65.2	127.4

Sol:- $y = ab^x$ —— (1)

$$\log_{10} y = \log_{10} a + x \log_{10} b$$

$$\Rightarrow \boxed{Y = A + BX} \text{ —— } (*)$$

where $Y = \log_{10} y$, $A = \log_{10} a$, $B = \log_{10} b$, $X = x$

corresponding normal eq for eq (*)

$$\boxed{\begin{aligned} \Sigma Y &= AN + B\Sigma X \quad -(2) \\ \Sigma XY &= A\Sigma X + B\Sigma X^2 \quad -(3) \end{aligned}}$$

$X = x$	y	X^2	$Y = \log_{10} y$	XY
2	8.3	4	0.919	1.82
3	15.4	9	1.187	3.54
4	33.1	16	1.519	6.04
5	65.2	25	1.814	9.07
6	127.4	36	2.105	12.63
$\Sigma X = 20$		$\Sigma X^2 = 90$	$\Sigma Y = 7.5$	$\Sigma XY =$ [illegible]

sub in 2 & 3, we get.

$N = 5,$

$7.5 = 5A + 20B$ — 3

$33.18 = 20A + 90B$ — 4

$\boxed{A = 0.31}$, $\boxed{B = 0.3}$

sub in $*$

~~[illegible]~~

$\rightarrow \boxed{a = 2.04}$, $\boxed{b = 1.995}$

from 2

$$\boxed{y = 2.04\,(1.995)^x}$$

4 fit $y = ab^x$ by the method of least square to the data given below.

x	0	1	2	3	4	5	6	7
y	10	21	35	59	92	200	400	610

A:- $(a = 10.499,\ b = 1.7959)$

$$y = ab^x \quad \text{—— (1)}$$

$$\log_{10} y = \log_{10} a + x \log_{10} b.$$

$X = x$	y	x^2	$Y = \log_{10} y$	XY
0	10	0	1	0
1	21	1	1.322	1.322
2	35	4	1.544	3.088
3	59	9	1.77	5.31
4	92	16	1.96	
5	200	25	2.301	
6	400	36	2.602	
	610	49	2.785	

from (1) $\boxed{y = (10.49)(1.79)^x}$

IV fitting of Power function :-

let $\boxed{y = ax^b}$ is the function to be fitted using the given data.

$\log_{10}$ O.B.S

$$\log_{10} y = \log_{10}(ax^b)$$

$$= \log_{10} a + \log_{10} x^b$$

$$\log_{10} y = \log_{10} a + b \cdot \log_{10} x$$

$$\boxed{Y = A + BX}$$

where . $Y = \log_{10}^{y}$, $A = \log_{10}^{a}$, $B = b$, $X = \log$

→ Power function can be fitted by using fitting of a st. line :

Problem :

(1) $y = ax^b$ to the following data.

x	1	2	3	4	5	6
y	2.98	4.26	5.21	6.10	6.8	7.5 .

Sol:- given curve is .

$$\boxed{y = ax^b} \longleftarrow (1)$$

$\log_{10}$. O.B.S

$$\log_{10}^{y} = \log_{10}^{a} + b \cdot \log_{10}^{x}$$

$$\Rightarrow \boxed{Y = A + BX} \longleftarrow (*)$$

normal eq :-

$$\Sigma y = AN + b\Sigma x \longrightarrow (2)$$

$$\Sigma xy = A\Sigma x + b\Sigma x^2 \longrightarrow (3)$$

x	y	x^2	$Y = \log_{10} y$	XY	$X = \log_{10} x$
1	2.98	0	0.474	0	0
2	4.26	0.09	0.629	0.186	0.3
3	5.21	0.22	0.716	0.333	0.47
4	6.10	0.36	0.785	0.468	0.6
5	6.8	0.47	0.832	0.57	0.69
6	7.5	0.59	0.875	0.664	0.77
		$\Sigma x^2 = 1.732$	$\Sigma Y = 4.218$		$\Sigma X = 2.83$

$\Sigma XY = 2.23$

② & ③ we get. $N=6$

$4.29 = 6A + 2.83B$ — ④

$2.23 = 2.83A + 1.7302B$ — ⑤

solving ④ & ⑤

$\boxed{A = 0.46}$, $B = 0.52$.

since :- $A = \log_{10} a$, $B = b = 0.52$

$\boxed{a = 2.88}$

from eq ① . required eq .

$y = ax^b$

$\Rightarrow \boxed{y = 2.88\, x^{0.52}}$

(2) fit a Power function of the form $y = ax^b$ from the data.

x	2	4	6	8	10
y	0.973	3.839	8.641	15.987	23.794

(3) fit a curve $y = ae^{bx}$ to the following data

x	1	5	7	9	12
y	10	15	12	15	21

(4) find $y = ab^x$ from the data.

x	0	1	3	4	5	6	7
y	10	21	59	92	202	400	610

(5) fit a 2nd degree curve to $y = a + bx + cx^2$

x	1	2	3	4	5	6	7	8	9
y	·2	6	7	8	10	11	11	10	9

sqn.

* Regression :-

the statistical method, which help us to estimate the unknown value of '1' variable from the known value of related variable is called Regression.

* Regression line :-

A st. line fitted to the data by the method of least square is called regression line.

Note :- there are two regression line constructed for the given data. one regression line shows the regression of "x on y" and other shows "y on x"

* Regression eq of Y on X :-

$y = a + bx$, where a, b are constant.

X on Y :-

$x = a + by$, where a, b are constant

normal eq for Y on X :-

$\Sigma y = aN + b\Sigma x$,

$\Sigma xy = a\Sigma x + b\Sigma x^2$.

X on Y :-

$$\Sigma X = aN + b\Sigma Y ,$$

$$\Sigma XY = a\Sigma Y + b\Sigma Y^2 .$$

Problem :-

(1) from the following data, obtain two regression line.

X	6	2	10	4	8
Y	9	11	5	8	7

Sol:- (i) Regression line for X on Y

$$\boxed{X = a + bY} \quad \text{— } (*)$$

normal eq :-

$$\Sigma X = aN + b\Sigma Y \quad \text{--- } (1)$$

$$\Sigma XY = a\Sigma Y + b\Sigma Y^2 \quad \text{--- } (2)$$

X	Y	XY	X^2	Y^2
6	9	54	36	81
2	11	22	4	121
10	5	50	100	25
4	8	32	16	64
8	7	56	64	49
$\Sigma X = 30$	$\Sigma Y = 40$	$\Sigma XY = 214$	$\Sigma X^2 = 220$	$\Sigma Y^2 = 340$

$\boxed{N = 5}$

solv 1 & 2

$(30) = a(5) + b(40)$ — 3

$(214) = 40a + 340b$ — 4

3 & 4

$\boxed{a = 16.4}$, $\boxed{b = -1.3}$

$\Rightarrow \boxed{x = 16.4 - 1.3y}$

(ii) y on x:-

$\boxed{N = 5}$

~~(30) = 5(a) + (40)~~

$(40) = a(5) + b(30)$

$(214) = a(30) + b(220)$

$\Rightarrow a = 11.9$, $b = 0.65$

$y = a + bx \Rightarrow \boxed{y = (11.9) - 0.65x}$

(2) from following data obtain two

(i)

x	2	4	6	8	10	12	14
y	4	2	5	10	4	11	12

(ii)

x	10	12	13	16	17	20	25
y	10	22	24	27	29	33	37

* Deviation taken from arithmetic mean of x and y :-

→ Regression eq of x on y is given by

$$\boxed{x-\bar{x} = r\frac{\sigma_x}{\sigma_y}(y-\bar{y})} \text{ — } x \text{ on } y.$$

$$\boxed{y-\bar{y} = r\cdot\frac{\sigma_y}{\sigma_x}(x-\bar{x})} \rightarrow y \text{ on } x.$$

Note : (1) regression coefficient $\underline{Y \text{ on } X}$ is

$$b_{yx} = r \cdot \frac{\sigma_y}{\sigma_x}$$

$\underline{X \text{ on } Y}$:-

$$b_{xy} = r \frac{\sigma_x}{\sigma_y}$$

(2) $b_{xy} \cdot b_{yx} = r \cdot \frac{\cancel{\sigma_x}}{\cancel{\sigma_y}} \times r \frac{\cancel{\sigma_y}}{\cancel{\sigma_x}}$

$$\Rightarrow \boxed{r^2 = b_{xy} \cdot b_{yx}}$$

$$\Rightarrow \boxed{r = \sqrt{b_{xy} \cdot b_{yx}}}$$

(3) Both the regression line Passing through $(\bar{x}, \bar{y})$ i.e. mean values obtain as the Point of intersection of regression line.

(4) $-1 \leq r \leq 1$ and $0 \leq r^2 \leq 1$
where r is coefficient of regression.

*Angle b/w two regression lines :-
let the regression line of 'x on y' and 'y on x' are

$$x - \bar{x} = r \frac{\sigma_x}{\sigma_y}(y - \bar{y}) \quad (\text{"x on y"})$$

$$\Rightarrow y - \bar{y} = \frac{1}{r} \frac{\sigma_y}{\sigma_x}(x - \bar{x}) \quad \text{—— (1)}$$

$$\Rightarrow y - \bar{y} = r \cdot \frac{\sigma_y}{\sigma_x}(x - \bar{x}) \quad \text{--- (2)} \quad (y \text{ on } x")$$

from (1) & (2),

let $m_1 = \frac{1}{r} \cdot \frac{\sigma_y}{\sigma_x}$ and $m_2 = r \cdot \frac{\sigma_y}{\sigma_x}$

$$\therefore \tan\theta = \frac{m_1 - m_2}{1 + m_1 m_2}$$

$$\Rightarrow \boxed{\tan\theta = \left(\frac{1 - r^2}{r}\right) \frac{\sigma_x \cdot \sigma_y}{\sigma_x^2 + \sigma_y^2}} \ *$$

Problem:-

① If θ is angle b/w two regression line and S.D of "Y" twice the S.D of "X" and $r=0.25$ then find $\tan\theta$.

Sol:- given that

$r = 0.25,$

$\sigma_y = 2\sigma_x$

$\therefore \tan\theta = \left(\frac{1-r^2}{r}\right)\frac{\sigma_x \cdot \sigma_y}{\sigma_x^2 + \sigma_y^2} = \underline{1.5}$

② If $\sigma_x = \sigma_y = \alpha$ and the angle b/w regression line $\tan^{-1}(4/3)$.

Sol:- given that,

$\theta = \tan^{-1}(4/3 \Rightarrow \tan\theta = 4/3$

$$4/3 = \left(\frac{1-r^2}{r}\right)\frac{\sigma_x \cdot \sigma_y}{\sigma_x^2 + \sigma_y^2}$$

$$\Rightarrow 4/3 = \left(\frac{1-r^2}{r}\right)\frac{\cancel{\sigma^2}}{2\cancel{\sigma^2}} \quad (\because \sigma_x = \sigma_y = \sigma)$$

$$\Rightarrow 8/3 = \left(\frac{1-r^2}{r}\right)$$

$$\Rightarrow \boxed{r = 1/3 = 0.33}$$

(3) the tangent of angle b/w two regression line 0.6 and if $\sigma_x = \frac{1}{2}\sigma_y$, find r.

Ans

sol:- $0.6 = \left(\frac{1-r^2}{r}\right) \cdot \frac{\sigma_x \cdot \sigma_y}{\sigma_x^2 + \sigma_y^2}$

$r = 0.5$

*** (4) find the correlation coefficient b/w x and y when the line of regression are $2x - 9y + 6 = 0$ & $x - 2y + 1 = 0$.

sol:- let the line of regression of x and y:

$\Rightarrow 2x = 9y - 6$ — (i)

$\Rightarrow x = 9/2\, y - 3$

let the line of regression of y on x is

$$\Rightarrow x - 2y + 1 = 0$$
$$y = \frac{1}{2}x + \frac{1}{2} \quad \text{—} \quad (2)$$

the regression coefficient

$$b_{xy} = 9/2 \quad , \quad b_{yx} = 1/2$$

$$r = \sqrt{b_{xy} \times b_{yx}} = \sqrt{\frac{9}{2} \times \frac{1}{2}} = 3/2 > 1 .$$

So, the choice of regression of line is incorrect

let us assume line of regression of

$\underline{x \text{ on } y}$ is.

$$\Rightarrow x - 2y + 1 = 0$$
$$\Rightarrow x = 2y - 1 \quad \text{—} \quad (3)$$

$\underline{y \text{ on } x}$ · is $\Rightarrow 2x - 9y + 6 = 0$

$$\Rightarrow y = 2/9\, x + 6/9 \quad \text{—} \quad (4)$$

Here, $b_{xy} = 2$, $b_{yx} = 2/9$

$\therefore r = \sqrt{b_{xy} \times b_{yx}} = \sqrt{2 \times \frac{2}{9}} = \frac{2}{3} < 1$

$\therefore$ Hence $\boxed{r = 2/3}$

(5) find the correlation coefficient of b/w x and y when the line of regression are $5y - 8x + 17 = 0$, $2y - 5x + 14 = 0$.

A:- $\boxed{r = .4/5}$

(6) The two regression eq's of the variables x and y

$x = 19.13 - 0.87y$ and $y = 11.64 - 0.50x$

find (i) mean of x.

(ii) mean of y.

(iii) correlation coefficient(r) b/w x and y

sol:- given regression eq are :-

$$\therefore x = 19.13 - 0.87y \quad \text{—} \ (1)$$

$$y = 11.64 - 0.50x \quad \text{—} \ (2)$$

Here, $b_{xy} = -0.87$, $b_{yx} = -0.50$

→ the two regression line passing through the mean $(\bar{x}, \bar{y})$ eq (1) & (2)

$$\therefore \bar{x} = 19.13 - 0.87\bar{y} \quad \text{—} \ (3)$$

$$\bar{y} = 11.64 - 0.50\bar{x} \quad \text{—} \ (4)$$

solving (3) & (4) we get:

$$\boxed{\bar{x} = 15.937} \quad \boxed{\bar{y} = 3.67}$$

(iii) $r = \sqrt{b_{xy} \times b_{yx}} = \sqrt{-0.87 \times 0.50} = 0.65$

$$\boxed{r = 0.65}$$

* Multiple linear regression (MLR):-

Consider "MLR"

$$y = a_0 + a_1 x_1 + a_2 x_2$$

where a_0, a_1, a_2 are constant.

$$\Sigma y = a_0 N + a_1 \Sigma x_1 + a_2 \Sigma x_2 \quad \text{—— (1)}$$

$$\Sigma x_1 y = a_0 \Sigma x_1 + a_1 \Sigma x_1^2 + a_2 \Sigma x_1 x_2 \quad \text{—— (2)}$$

$$\Sigma x_2 y = a_0 \Sigma x_2 + a_1 \Sigma x_1 x_2 + a_2 \Sigma x_2^2 \quad \text{—— (3)}$$

Problem :-

(1) find the least square regression eq y on x_1 and x_2 from the following data.

y	3	5	6	8	12	14
x_1	16	10	7	4	3	2
x_2	90	72	54	42	30	12

Sol:- let $Y = a_0 + a_1 x_1 + a_2 x_2$ —— (*)

req multiple eq :-

corresponding normal eq

$$\Sigma y = a_0 N + a_1 \Sigma x_1 + a_2 \Sigma x_2 \quad \text{--- (1)}$$

$$\Sigma x_1 y = a_0 \Sigma x_1 + a_1 \Sigma x_1^2 + a_2 \Sigma x_1 x_2 \quad \text{--- (2)}$$

$$\Sigma x_2 y = a_0 \Sigma x_2 + a_1 \Sigma x_1 x_2 + a_2 \Sigma x_2^2 \quad \text{--- (3)}$$

x_1	x_1^2	x_2	x_2^2	y	$x_1 y$	$x_2 y$	$x_1 x_2$
16	256	90	8100	3	48	270	1440
10	100	72	5184	5	50	360	720
7	49	54	2916	6	42	324	378
4	16	42	1764	8	32	336	168
3	9	30	900	12	36	360	90
2	4	12	144	14	28	168	24
$\Sigma x_1 =$ 42	$\Sigma x_1^2 =$ 434	$\Sigma x_2 =$ 300	$\Sigma x_2^2 =$ 19008	$\Sigma y =$ 48	$\Sigma x_1 y =$ 236	$\Sigma x_2 y =$ 1818	$\Sigma x_1 x_2 =$ 2820

Here $N = 6$

sub in eq (1), (2), (3) we get.

$$48 = 6a_0 + 42a_1 + 300a_2 \quad \text{--- (4)}$$

$$236 = 42a_0 + 434a_1 + 19008a_2 \quad \text{--- (5)}$$

$$1818 = 300a_0 + 2820a_1 + 19008a_2 \quad \text{--- (6)}$$

solving (4), (5), (6) we get.

$$a_0 = 16.1, \ a_1 = 0.42, \ a_2 = -0.22$$

required eq^n:-

(*) $\therefore$ $\boxed{y = 16.1 + 0.42x_1 - 0.22x_2}$

(2) the following data on the no. of twists required to break the certain kind of forged along a ball present in the metal.

No. of twists (y)	41	49	69	65	40	50	58	57	31	36	44
Percentage A (x_1)	1	2	3	4	1	2	3	4	1	2	3
element B (x_2)	5	5	5	5	10	10	10	10	15	15	15

No. of twists (y)	57	19	31	33	43
Percentage A (x_1)	4	1	2	3	4
element B (x_2)	15	20	20	20	20

fit a least square regression and use its eqn to estimate the no. of twists req to break one of the balls $x_1 = 2.5$ and $x_2 = 12$

x_1	x_1^2	x_2	x_2^2	y	x_1y	x_2y	x_1x_2
1	1	5	25	41	41	210	5
2	4	5	25	49	98	245	10
3	9	5	25	69	207	345	15
4	16	5	25	65	260	325	20
1	1	10	100	40	40	400	10
2	4	10	100	50	100	500	20
3	9	10	100	58	174	580	30
4	16	10	100	57	228	570	40
1	1	15	225	31	31	465	15
2	4	15	225	36	72	540	30
3	9	15	225	44	132	660	45
4	16	15	225	57	228	855	60
1	1	20	400	19	19	380	20
2	4	20	400	31	62	620	40
3	9	20	400	33	99	660	60
4	16	20	400	43	172	860	80

$\Sigma x_1 = 40$

$\Sigma x_1^2 = 120$ $\Sigma y = 723$ $N = 16$

$\Sigma x_2 = 200$ $\Sigma x_1 y = 1963$

$\Sigma x_2^2 = 3000$ $\Sigma x_2 y = 8215$

$\Sigma x_1 x_2 = 500$

$$723 = 16a_0 + 40a_1 + 200a_2 \quad \text{--- } (1)$$

$$1963 = 40a_0 + 120a_1 + 500a_2 \quad \text{--- } (2)$$

$$8215 = 200a_0 + 500a_1 + 3000a_2 \quad \text{--- } (3)$$

$$a_0 = 46.3,\ a_1 = 7.7,\ a_2 = -1.64$$

$$\boxed{y = a_0 + a_1x_1 + a_2x_2} \quad \text{--- (a)}$$

$$y = 46.3 + 7.7(2.5) + (-1.64)(12)$$

$$y = 46.3 + 19.25 - 19.68$$

$$\therefore \boxed{y = 45.87.}$$

for "3" Problems.

$$\boxed{\begin{aligned} z &= a_0 + a_1x + a_2y \\ \Sigma z &= a_0N + a_1\Sigma x + a_2\Sigma y \\ \Sigma xz &= a_0\Sigma x + a_1\Sigma x^2 + a_2\Sigma xy \\ \Sigma yz &= a_0\Sigma y + a_1\Sigma xy + a_2\Sigma y^2. \end{aligned}}$$

Rev-Problem :-

① if two regression coefficient r 0.8 & 0.2 what would be the value of coefficient of correlation.

sol:- let $b_{xy} = 0.8$, $b_{yx} = 0.2$

$$r = \sqrt{b_{xy} \cdot b_{yx}} = \sqrt{0.8 \times 0.22} = \underline{0.4}.$$

② the regression line are $7x - 16y + 9 = 0$, $5y - 4x - 3 = 0$, find $\bar{x}, \bar{y}$ and r.

sol:- let ~~the~~ regression eq $\underline{x \text{ on } y}$ is

$$7x - 16y + 9 = 0$$

$$\Rightarrow x = \frac{16}{7} y - 9/7$$

$\underline{y \text{ on } x}$ is $5y - 4x - 3 = 0$

$$\Rightarrow y = \frac{4}{5}x + \frac{3}{5}$$

let $b_{xy} = 16/7$, $b_{yx} = 4/5$

$$r = \sqrt{b_{xy} \times b_{yx}} = \sqrt{\frac{16}{7} \times \frac{4}{5}} = 1.35 \times$$

As $-1 \le r \le 1$ the of regression

eq is incorrect.

let us assume $\underline{x \text{ on } y}$ is

$$5y - 4x - 3 = 0$$

$$\Rightarrow x = \frac{5}{4}y - \frac{3}{4}$$

y on x is $7x - 16y + 9 = 0$

$$\Rightarrow y = \frac{7}{16}x + \frac{9}{16}$$

$b_{xy} = 5/4$, $b_{yx} = 7/16$

$$\boxed{r = \sqrt{\frac{5}{4} \times \frac{7}{16}} = 0.73}$$

the mean $\bar{x}$ and $\bar{y}$ passes through the given regression line.

$\therefore 7\bar{x} - 16\bar{y} + 9 = 0$,

$-4\bar{x} + 5\bar{y} - 3 = 0$, solving.

$\boxed{\bar{x} = -0.103}$ $\boxed{\bar{y} = 0.577}$

3) find the regression line and correlation coefficient :-

(a)

X	1	3	5	7	8	10
Y	8	12	15	17	18	20

(b)

X	40	34	28	30	44	[illegible]	31
Y	30	31	26	29	32	33	28

sol: a regression line of x on y

$$\boxed{X = a + bY} \longrightarrow (*)$$

normal eq:-

$$\Sigma X = aN + b\Sigma Y \longrightarrow (1)$$

$$\Sigma XY = a\Sigma X + b\Sigma Y^2 \longrightarrow (2)$$

X	X^2	Y	Y^2	XY
1	1	8	64	8
3	9	12	144	36
5	25	15	225	75
7	49	17	289	119
8	64	18	324	144
10	100	20	400	200

$N=6$ $\Sigma x = 34$ $\Sigma y = 90$ $\Sigma xy = 58$

$\Sigma y^2 = 1446$

$\Sigma x^2 = 248$

$34 = 6a + 90b$ — (3)

$582 = 34a + 1446b$ — (4)

$\boxed{a = -5.58, b = 0.75}$

Y on x :-

Normal :-

$\Sigma y = aN + b\Sigma x$

$\Sigma xy = a\Sigma x + b\Sigma x^2$

$90 = 6N + 34b$ — (1)

$582 = 34a + 248b$ — (2)

$a = 7.62, b = 1.30$

X on Y :-

$$X = -5.58 + 0.75Y$$

Y on X :-

$$Y = 7.62 + 1.3X$$

$$r = \sqrt{b_x \times b_y} = \sqrt{1.3 \times 0.75}$$

$$\boxed{r = 0.98}$$

(b) given that,

x	y	xy	x^2	y^2
40	30	1200	1600	900
34	31	1054	1156	961
28	26	728	784	676
30	29	870	900	841
44	32	1408	1936	1024
38	33	1254	1444	1089
31	28	868	961	784.

$\Sigma x = 245$

$\Sigma y = 209$

$\Sigma xy = 7382$ $\quad N = 7.$

$\Sigma x^2 = 8781$

$\Sigma y^2 = 6275.$

X on Y :-

$245 = 7a + 209b$ — (1)

$7382 = 245a + 6275b$ — (2)

$a = 0.75, \ b = 1.14$

$$\boxed{X = 0.75 + 1.14Y}$$

y on x :-

$$209 = 7a + 245b \quad \text{--- } (3)$$

$$7382 = 245a + 8781 \cdot b \quad \text{--- } (4)$$

$$a = 18.4 \quad , b = 0.32$$

$$\boxed{y = 18.4 + 0.32x}$$

$$r = \sqrt{b_x \times b_y} = \sqrt{1.14 \times 0.32}$$

$$\boxed{r = 0.603}$$

Thank you